Ruchitha Gajawada

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Professional Experience

Apziva, *AI Resident* — July 2024 – present | Beaumont, USA

- Developed a predictive model for customer satisfaction classification using data preprocessing, feature engineering, and Machine Learning algorithms. Enhanced predictions with Exploratory Data Analysis (EDA) and Lazy Predict.
- Implemented an ensemble Voting Classifier (Hard Voting) with Logistic Regression, SVM, and Decision Trees, achieving a recall of 91% for unhappy customers.
- Optimized model performance through hyperparameter tuning, improving computational efficiency and reducing resource usage by 25%. Applied dimensionality reduction to maintain high recall (Class 1: 91%, Class 0: 67%).
- Enhanced operational efficiency by reducing false negatives and delivering a robust ML model for proactive customer dissatisfaction detection.

Lamar University, *Graduate Teaching Assistant - Data Science* — September 2023 – May 2024 | Beaumont, USA

- Developed a machine learning web app using Python Flask to predict student performance, utilizing features like gender, race, and parental education for accuracy.
- Managed version control with GitHub, created development environments, and handled dependencies with setup.py and requirements.txt for effective project collaboration in Visual Studio Code.
- Built a scalable data pipeline in Python for ingestion, transformation, and model training. Used hyperparameter optimization, achieving high performance with Linear Regression (R^2 = 87.91%) and Random Forest Regressor (R^2 = 85.06%).
- Deployed the solution on AWS Elastic Beanstalk for scalability and ease of access.

Skills

Programming Languages: C, Python, SQL, Java, R

Data Manipulation and Visualisation:: Pandas, NumPy, Power BI, Matplotlib, Seaborn.

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, Decision Trees, Random Forest, Regression, Classification, Clustering, Deep Learning, Natural Language Processing, Computer Vision

Databases: MySQL, SQL Server

Cloud Technologies: AWS, AWS EC2, AWS Lambda, AWS IAM

Tools and Frameworks: Jupyter Notebook, Git, GitHub, Visual Studio Code, Flask, Streamlit, Django

Projects

Term Deposit Marketing — July 2024 – September 2024

[Machine Learning, pycaret,automl,optuna]

- Developed a predictive model to optimize marketing strategies for bank term deposits using Python, sklearn, and imbalanced-learn for data processing and model training.
- Applied SMOTE-ENN, PyCaret, and AutoML TPOT for balanced dataset creation and model selection, achieving up to 92% accuracy and significant recall improvements in minority class prediction.
- Performed customer segmentation to tailor marketing efforts, utilizing feature importance analysis and SHAP values to reduce unwanted calls by 96% while maintaining high precision and recall rates.

Email Spam Detection Using Classification — January 2024 – May 2024

[Naive Bayes, NLTK]

- Developed a spam email classifier with NLP and ML using NLTK for preprocessing and Naive Bayes for classification.
- Leveraged TF-IDF vectorization for feature extraction and deployed on AWS with Flask, configured an EC2 instance, and utilized serverless computing for load balancing and scalability.
- Achieved 97% accuracy and 100% precision on test data, addressing class imbalances with Bayesian inference for optimal performance.

Content-based Movie Recommendation System — September 2023 – January 2024

[Python, Recommendation System, AWS]

- Developed a movie recommendation system using NLP and ML, employing Bag of Words (Bow) and Cosine Similarity for over 4,800 movies. Achieved a 15% increase in recommendation accuracy.
- Implemented data preprocessing with stop word removal and stemming, integrating genres, keywords, cast, and crew information to enhance content representation.
- Deployed on AWS with Flask and PyCharm, utilized Pickle for model serialization, and integrated TMDB API. Configured AWS EC2 with CI/CD pipelines for continuous integration and real-time interactions.

Education

Master of Science in Computer Science, *Lamar University* — August 2022 – May 2024 | Beaumont, USA